

Stack Overflow survey Analysis

Sector: Technology / Human Resources

Team :- kuch bhi

Team member:-

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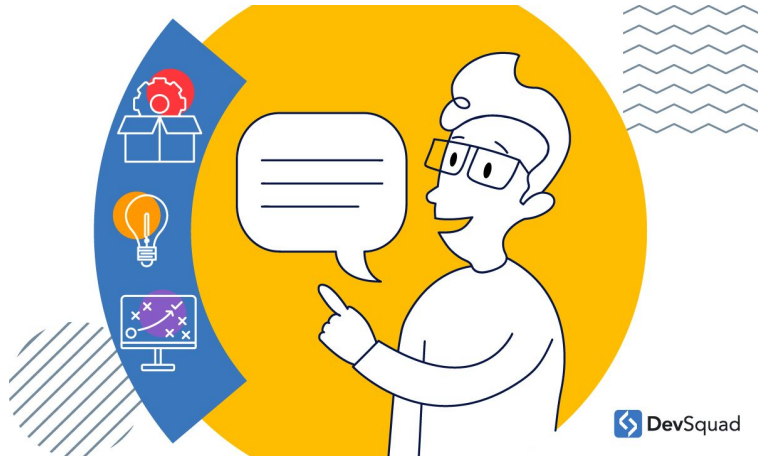
Context and Problem Statement

Sector Context: Rapid tech changes → unclear salary benchmarks & evolving skills.

Stakeholders: Recruiters, Hiring Managers, Job Seekers.

Core Question: What drives developer pay & impact of remote work + AI?

Objective: Build a dashboard to analyze salaries, tech trends, and work preferences.



Data Engineering

Source: Stack Overflow Developer Survey 2024

Size & Coverage: ~60,000 raw responses, filtered to ~33,000 valid global entries

Data Cleaning: Removed rows with missing **ConvertedCompYearly**, filled missing values (mode for categorical, median for **WorkExp**), and trimmed top & bottom 1% salary outliers.

Key Fields: **ConvertedCompYearly**, **WorkExp**, **RemoteWork**,

Primary_Language, **AISelect**, and **JobSat**



KPI Framework

Avg Salary: Average annual compensation (USD); useful for benchmarking market rates.

Job Satisfaction: Self-reported (out of 10); reflects industry health and retention risk.

AI Adoption: % of developers using AI tools; indicates future skill trends.

Avg Salary
\$21,776

Job Satisfaction
6.429 / 10

AI Adoption
100.0 %

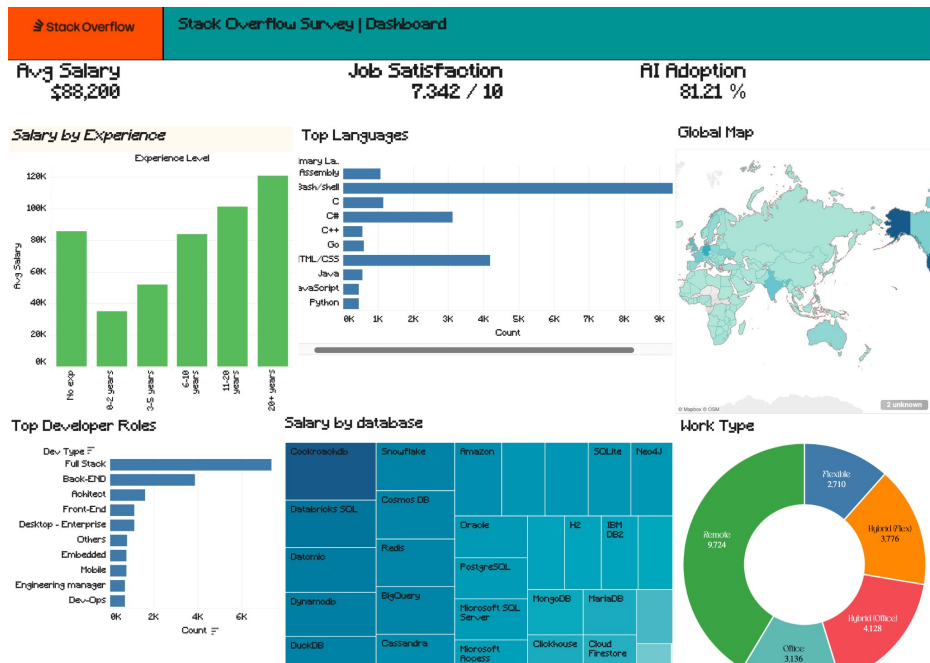
Key EDA Insights

- **Education Premium:** Higher formal degrees correlate with higher median salaries globally.
- **Experience Matters:** Salaries increase consistently with years of professional coding experience.
- **Tech Stack:** Python and JavaScript remain the most widely used primary languages.
- **Remote Dominance:** A significant share of developers work fully remotely or in hybrid setups.
- **AI Integration:** Over 90% of developers use or plan to use AI tools in their workflow.



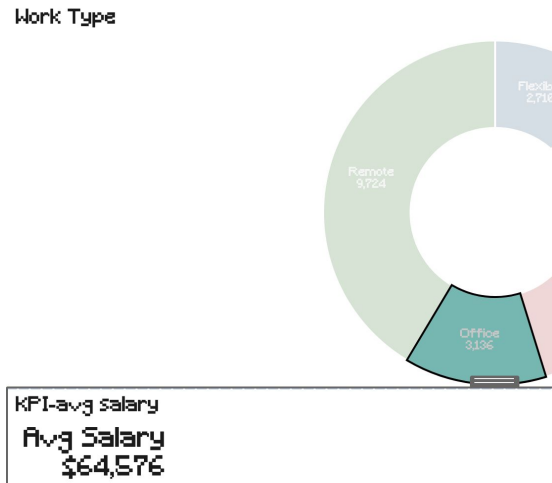
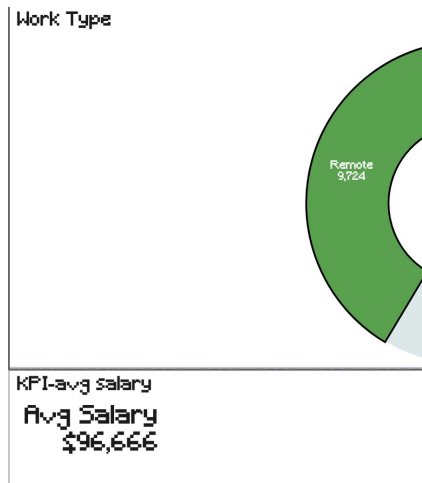
DASHBOARD

- **Executive View:** High-level KPIs (Avg Salary, Job Satisfaction, AI Adoption).
- **Operational View:** Salary breakdown by Database, Language, and Experience level.
- **Filters & Drilldowns:** Interactive filters for Primary Language, Country (map), and Developer Role.



Top Insights

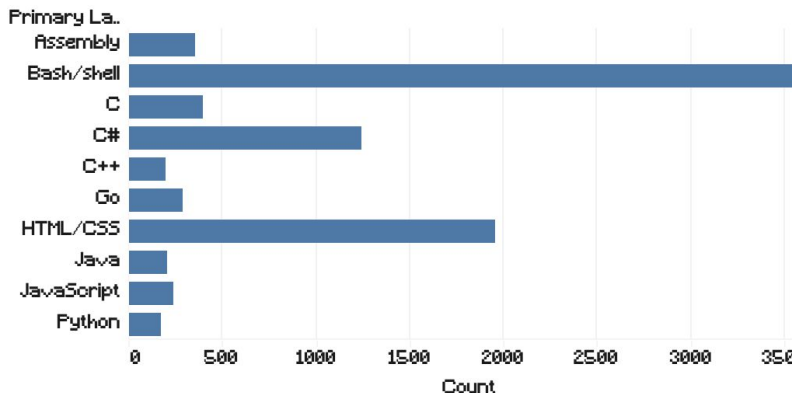
1. Remote Work Pays More: Remote roles consistently command higher compensation packages across major markets.
2. AI is Non-Negotiable: With massive adoption rates, AI competency is becoming a baseline expectation, not a luxury.
3. Core Languages Drive Volume: While niche languages exist, mastering Python and JavaScript opens the most opportunities.



Recommendations

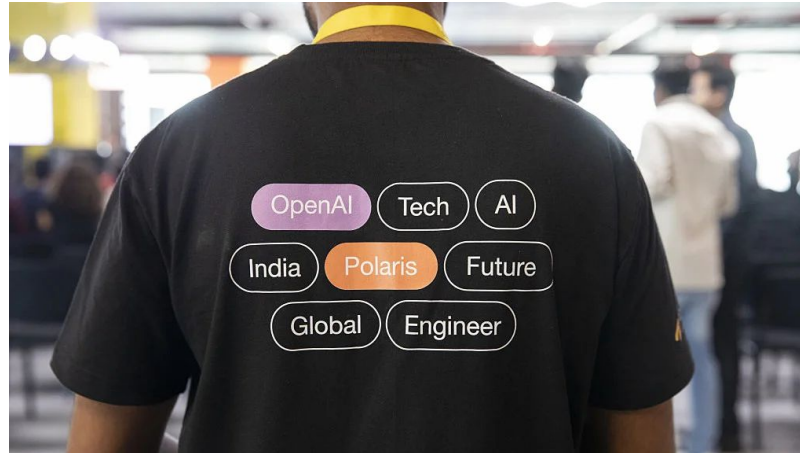
1. Adopt Remote Policies: Companies should offer remote or highly flexible hybrid options to attract top talent and remain competitive on salary.
2. Standardize AI Training: Organizations must provide official AI tools (like GitHub Copilot) and training to keep pace with developer workflows.
3. Focus Hiring on Core Stacks: Recruiters should prioritize sourcing talent proficient in JavaScript and Python to fill full-stack and backend roles efficiently.

Top Languages



Impact

- Expected Outcome: Improved talent acquisition (by offering remote roles) and increased developer productivity (via AI integration).
- Priority and Feasibility: High Priority, High Feasibility. Implementing remote policies and AI subscriptions requires low technical overhead but high policy alignment.



Limitations

- Data Constraints: Survey data is self-reported, which can lead to biases in stated salaries and job satisfaction.
- Method Constraints: Compensation is converted to USD, which may skew comparisons due to purchasing power parity and local cost of living differences.

